

Rajendran Prathibha Devkar

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Software Engineer with 4+ years building ML systems and full-stack applications across enterprise analytics and independent research. Specializes in deep learning architecture design and applied LLM systems evaluation across PyTorch, Ollama, and React/Node.js, with co-authored published research in gesture recognition.

EDUCATION

Master of Science in Computer Science — The University of Texas at Arlington, USA May 2026

Bachelor of Technology in Computer Science — Saintgits College of Engineering, India May 2019

EXPERIENCE

Research Assistant, SEAR Lab — The University of Texas at Arlington Apr 2026 - Present

- Refactored a 6-service Docker stack (Mosquitto MQTT, InfluxDB, Grafana, a FastAPI Data Hub, Python collectors) into a single self-contained FastAPI application with an embedded dashboard, eliminating the Docker/Grafana/MQTT infrastructure dependency entirely.
- Integrated live telemetry from three hardware APIs — LG ThinQ (AC), Rheem EcoNet (water heater), Enphase (solar) — into a unified REST API (/api/data, /api/history, /api/status) computing a real-time Net Load metric (solar minus HVAC/HWH draw), via a background async collector refreshing every 60 seconds.

Software Developer Intern, Board Americas — Remote, Texas May 2025 - Mar 2026

- Resolved 45+ medium-to-high severity production bugs over 11 months across an enterprise forecasting platform (Aurelia/TypeScript, C#), improving platform stability across UI, state management, and backend API layers.
- Diagnosed and fixed critical D3.js charting bugs, including fiscal-year x-axis display errors and incorrect actual-vs-forecast rendering, correcting chart data clients relied on for business decisions.
- Fixed a company-wide cache-invalidation bug that let forecast data drift out of sync across users, by changing invalidation scope from per-user to per-company; also resolved a null-date bug across all datepicker components that was triggering 500 errors in production.

Technology Analyst, Infosys Private Ltd. — Trivandrum, India Oct 2019 - Nov 2022

- Developed React Native UI and integrated backend APIs for a curbside-pickup feature added to the Tractor Supply Company mobile app, improving user satisfaction by 30% based on pilot-release feedback.
- Built RESTful APIs in Node.js for an e-learning platform supporting 200+ concurrent users across student, teacher, and admin roles, enabling real-time session scheduling, participation, and new-user registration.
- Designed PostgreSQL schemas and optimized SQL queries for the platform; built a JMeter load-testing pipeline simulating 1K-10K virtual users, which identified and resolved a calendar API bottleneck for a team-reported 40% improvement in data-retrieval speed.
- Deployed the platform to live servers via PuTTY/WinSCP and managed process reliability with PM2, ensuring stable delivery across environments.

SKILLS

ML: PyTorch, TensorFlow, Keras, OpenCV, NumPy, Pandas, CNN

AI/LLM: Ollama (local LLM inference), Prompt Engineering, DistilBERT fine-tuning, LLM evaluation & confidence calibration

Databases: SQL Server, PostgreSQL, MySQL

Frontend: React, React Native, Aurelia, D3.js, HTML5, CSS3

Languages: Python, TypeScript, JavaScript, C#

Backend / APIs: ASP.NET, Node.js, FastAPI, RESTful APIs

Cloud: AWS, GCP

DevOps / Tools: Azure DevOps, Git/GitHub, Jira, Docker

PROJECTS

Recurrent Depth Transformer (RDT) — PyTorch, Python

- Implemented a Recurrent Depth Transformer from scratch in PyTorch, replacing stacked transformer layers with a single recurrently-looped block; matched a parameter-matched GPT baseline's accuracy on multi-hop reasoning (1-5 hops) at 30-47% fewer parameters depending on configuration, then ran inference-time loop-scaling experiments (6/10/14 loops) testing generalization to unseen chain lengths.
- Built and evaluated a tool-calling agent (3 tools, 17 test tasks) from the fine-tuned RDT model; caught a misleading evaluation metric that reported 100% task completion regardless of correctness, manually verified only 1-of-4 single-step answers were factually correct, and rewrote the evaluation harness to report results honestly instead of overstating them.

Confidence-Aware Meeting Intelligence — Python, Ollama, DistilBERT, Streamlit

- Built a confidence-aware meeting summarizer and action-item extractor that scores the reliability of each output instead of presenting everything with equal certainty, unlike Otter.ai/Fireflies-style tools (60-90% accuracy, no confidence signal): a local LLM (Ollama, llama3.2:3b) extracts summaries and action items with owner, due date, and a supporting quote, independently cross-checked by a separately fine-tuned DistilBERT classifier.

- Calibrated a confidence score (logistic regression on quote-grounding, owner-attribution plausibility, cross-model agreement, and self-reported confidence) on 228 hand-labeled examples; held-out accuracy: 58% for high-confidence items vs. 16% for low-confidence, a 3.6x spread confirming a genuine trust signal.
- Extracted the confidence-calibration logic into confidence-referee, a standalone MIT-licensed Python package (13 passing tests, CI across Python 3.9-3.12) that this project now depends on directly; validated it generalizes to a second, unrelated domain — RAG-answer hallucination detection on HaluEval — where its grounding signal alone reached 0.97 AUC and 94% held-out accuracy.

Apex F1 Predictor — *Python, FastAPI, scikit-learn, WebSocket*

- Built Apex F1 Predictor, a full-stack ML app forecasting Formula 1 race outcomes from historical race data (2023+) with a RandomForestClassifier and a FastAPI backend (autonomous retraining after each race, WebSocket-based live telemetry); reached AUC of 0.97/0.91/0.82 for win/podium/points-finish probability, though exact finishing-position prediction is harder (~4-position MAE) — a limitation surfaced directly in the app rather than hidden.

PUBLICATION

Devkar, R. Prathibha, Daniel, M.S., John, N.P., et al. (2021). Speaking Mouth System for Speech-Impaired People using Hand Gestures

Built a custom 4,000-image dataset (2,400 train / 1,600 test), architected a 2-layer CNN achieving 96% test accuracy, and implemented an end-to-end inference pipeline (gesture → text → synthesized speech); published in Springer FSST 2019.